

Fundamental field	Symmetries	Decomposition into SO(3) irrep(s)	Source
$C_{\alpha\beta\chi}$	$C_{\chi\beta\alpha} == - C_{\alpha\beta\chi}$	$-\frac{1}{2} \eta_{\beta\chi} C_{1^- \alpha}^{\#1} + \frac{1}{2} \eta_{\alpha\beta} C_{1^- \chi}^{\#1} + \frac{4}{3} C_{2^- \alpha\chi\beta}^{\#1} + \frac{1}{3} \eta_{\beta\chi} C_{0^+}^{\#1} n_\alpha - C_{1^+ \beta\chi}^{\#1} n_\alpha + C_{2^+ \beta\chi}^{\#1} n_\alpha + C_{1^+ \alpha\chi}^{\#2} n_\beta - \frac{1}{2} C_{1^- \chi}^{\#1} n_\alpha n_\beta - C_{1^- \chi}^{\#2} n_\alpha n_\beta - \frac{1}{3} \eta_{\alpha\beta} C_{0^+}^{\#1} n_\chi - C_{1^+ \alpha\beta}^{\#1} n_\chi - C_{2^+ \alpha\beta}^{\#1} n_\chi + \frac{1}{2} C_{1^- \alpha}^{\#1} n_\beta n_\chi + C_{1^- \alpha}^{\#2} n_\beta n_\chi + \frac{1}{6} \epsilon \eta_{\alpha\beta\chi\delta} C_{0^-}^{\#1} n^\delta$	$J_{\alpha\beta\chi}$
SO(3) irrep	Symmetries	Expansion in terms of the fundamental field	Source SO(3) irrep
$C_{0^+}^{\#1}$	(None)	$C_{\alpha \beta}^{\beta} n^\alpha$	$J_{0^+}^{\#1}$
$C_{0^-}^{\#1}$	(None)	$\epsilon \eta_{\alpha\beta\chi\delta} C^{\beta\chi\delta} n^\alpha$	$J_{0^-}^{\#1}$
$C_{1^+ \alpha\beta}^{\#1}$	$C_{1^+ \beta\alpha}^{\#1} == - C_{1^+ \alpha\beta}^{\#1}$	$-\frac{1}{2} C_{\alpha\beta\chi} n^\chi + \frac{1}{2} C_{\beta\alpha\chi} n^\chi - \frac{1}{2} C_{\beta\chi\delta} n_\alpha n^\chi n^\delta + \frac{1}{2} C_{\alpha\chi\delta} n_\beta n^\chi n^\delta$	$J_{1^+ \alpha\beta}^{\#1}$
$C_{1^+ \alpha\beta}^{\#2}$	$C_{1^+ \beta\alpha}^{\#2} == - C_{1^+ \alpha\beta}^{\#2}$	$C_{\alpha\chi\beta} n^\chi + C_{\beta\chi\delta} n_\alpha n^\chi n^\delta - C_{\alpha\chi\delta} n_\beta n^\chi n^\delta$	$J_{1^+ \alpha\beta}^{\#2}$
$C_{1^- \alpha}^{\#1}$	(None)	$- C_{\alpha \beta}^{\beta} + C_{\beta \chi}^{\chi} n_\alpha n^\beta + C_{\alpha\beta\chi} n^\beta n^\chi$	$J_{1^- \alpha}^{\#1}$
$C_{1^- \alpha}^{\#2}$	(None)	$C_{\alpha\beta\chi} n^\beta n^\chi$	$J_{1^- \alpha}^{\#2}$
$C_{2^+ \alpha\beta}^{\#1}$	$C_{2^+ \beta\alpha}^{\#1} == C_{2^+ \alpha\beta}^{\#1}$	$-\frac{1}{2} C_{\alpha\beta\chi} n^\chi - \frac{1}{2} C_{\beta\alpha\chi} n^\chi - \frac{1}{3} \eta_{\alpha\beta} C_{\chi \delta}^{\delta} n^\chi + \frac{1}{3} C_{\chi \delta}^{\delta} n_\alpha n_\beta n^\chi + \frac{1}{2} C_{\beta\chi\delta} n_\alpha n^\chi n^\delta + \frac{1}{2} C_{\alpha\chi\delta} n_\beta n^\chi n^\delta$	$J_{2^+ \alpha\beta}^{\#1}$
$C_{2^- \alpha\beta\chi}^{\#1}$	$C_{2^- \beta\alpha\chi}^{\#1} == - C_{2^- \alpha\beta\chi}^{\#1}$	$\frac{1}{4} C_{\alpha\beta\chi} + \frac{1}{2} C_{\alpha\chi\beta} - \frac{3}{8} \eta_{\beta\chi} C_{\alpha \delta}^{\delta} - \frac{1}{4} C_{\beta\alpha\chi} + \frac{3}{8} \eta_{\alpha\chi} C_{\beta \delta}^{\delta} - \frac{3}{8} C_{\beta \delta}^{\delta} n_\alpha n_\chi + \frac{3}{8} C_{\alpha \delta}^{\delta} n_\beta n_\chi + \frac{1}{2} C_{\beta\chi\delta} n_\alpha n^\delta + \frac{1}{4} C_{\beta\delta\chi} n_\alpha n^\delta + \frac{1}{4} C_{\chi\beta\delta} n_\alpha n^\delta + \frac{3}{8} \eta_{\beta\chi} C_{\delta \epsilon}^{\epsilon} n_\alpha n^\delta - \frac{1}{2} C_{\alpha\chi\delta} n_\beta n^\delta - \frac{1}{4} C_{\alpha\delta\chi} n_\beta n^\delta - \frac{1}{4} C_{\chi\alpha\delta} n_\beta n^\delta - \frac{3}{8} \eta_{\alpha\chi} C_{\delta \epsilon}^{\epsilon} n_\beta n^\delta - \frac{1}{4} C_{\alpha\beta\delta} n_\chi n^\delta - \frac{1}{2} C_{\alpha\delta\beta} n_\chi n^\delta + \frac{1}{4} C_{\beta\alpha\delta} n_\chi n^\delta + \frac{3}{8} \eta_{\beta\chi} C_{\alpha\delta\epsilon} n^\delta n^\epsilon - \frac{3}{8} \eta_{\alpha\chi} C_{\beta\delta\epsilon} n^\delta n^\epsilon - \frac{3}{8} C_{\beta\delta\epsilon} n_\alpha n_\chi n^\delta n^\epsilon + \frac{3}{8} C_{\alpha\delta\epsilon} n_\beta n_\chi n^\delta n^\epsilon$	$J_{2^- \alpha\beta\chi}^{\#1}$